Create a Static Web Page Which Defines All Text

Formatting Tags of HTML in Tabular Format

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1. Objectives

- To understand the purpose and usage of all HTML text formatting tags across HTML 3.2 and HTML5.
- To create a well-structured static web page that presents all text formatting tags in a clear, readable tabular format.
- To differentiate between semantic tags (e.g., ,) and physical/presentational tags (e.g., <b>, <i>), and identify deprecated tags with their modern CSS alternatives.

2. Software / Hardware Requirements

Software Requirements:

- Operating System: Windows 10 / 11, macOS, or any Linux distribution
- Text / Code Editor: VS Code, Sublime Text, Notepad++, or any plain-text editor
- Web Browser: Google Chrome, Mozilla Firefox, Microsoft Edge (latest version)
- Language Used: HTML5 (no additional libraries or frameworks required)
- Server: Not required — file can be opened directly in browser as a local file

Hardware Requirements:

- Processor: Intel Core i3 or equivalent (any modern CPU)
- RAM: Minimum 2 GB (4 GB recommended)
- Storage: At least 10 MB free space
- Display: Standard monitor with a minimum resolution of 1024 x 768

3. Theory

3.1 Introduction to HTML

HTML (HyperText Markup Language) is the standard markup language used to create and structure content on the World Wide Web. Developed by Tim Berners-Lee in 1991, HTML has evolved through multiple versions — from HTML 1.0 to the current HTML5 — each version introducing new tags, attributes, and capabilities to support richer, more accessible, and more semantic web content.

An HTML document is composed of elements represented by tags enclosed in angle brackets (< >). Tags are usually paired — an opening tag such as and a closing tag such as — though some tags are self-closing (e.g.,
, <wbr>). Browsers interpret these tags and render the content visually on screen. HTML works alongside CSS (for styling) and JavaScript (for behaviour) to form the foundational trio of front-end web development.

3.2 Text Formatting in HTML

Text formatting tags in HTML are inline elements that modify the visual appearance or convey semantic meaning of text within a document. These tags are broadly categorised into the following groups:

- Semantic Tags: Carry meaningful information about the enclosed text. Examples include (important), (emphasis), <mark> (highlighted), <abbr> (abbreviation), <cite> (title of a work), <dfn> (definition), <time>, <data>, <q>, <blockquote>, <address>, <code>, <kbd>, <samp>, and <var>.
- Physical / Presentational Tags: Define the visual appearance of text without conveying meaning. Examples include <b> (bold), <i> (italic), <u> (underline), <s> (strikethrough), <small>, <sup>, <sub>.
- Structural / Utility Tags: Help organise or control inline text flow. Examples include , <pre>, <ins>, <del>, <bdi>, <bdo>, <wbr>.
- Ruby Annotation Tags: Used for East Asian typography to show pronunciation. Examples include <ruby>, <rt>, <rp>.
- Deprecated Tags: Tags removed in HTML5 because CSS provides equivalent functionality. Examples include <font>, <big>, , <strike>, <tt>.

3.3 Key Tags Learned and Used

The following are the important tags studied and implemented in this assignment:

<b> vs : Both render text in bold, but carries semantic importance (used for critical information), while <b> is purely presentational (used for keywords, product names).

<i> vs : Both italicise text, but adds stress emphasis that changes the meaning of a sentence. <i> is used for technical terms, foreign words, or thoughts.

<mark>: Introduced in HTML5, this tag highlights text with a yellow background by default, indicating relevance in the current context (e.g., search results).

<code>, <kbd>, <samp>, <var>, <pre>: A family of computer-related tags. <code> marks inline code; <kbd> denotes keyboard input; <samp> shows sample output; <var> marks variables; <pre> preserves whitespace and line breaks.

<sup> and <sub>: Used for superscript ($E=mc^2$) and subscript (H_2O), common in scientific and mathematical content.

<ins> and <del>: Represent editorial changes to a document — inserted and deleted text respectively. Both support cite and datetime attributes for traceability.

<abbr>: Wraps abbreviations and uses the title attribute to display the full form on hover, improving accessibility.

<bdi> and <bdo>: Used in multilingual pages. <bdi> isolates a text segment from its surroundings' directionality; <bdo> explicitly overrides the direction.

3.4 HTML Text Formatting Tags — Reference Table

#	Tag	Description	Example / Output
1	<b>	Bold text – no semantic meaning	Bold Text
2		Important text – semantically bold	Strong Text
3	<i>	Italic text – no semantic meaning	Italic Text
4		Emphasized text – semantically italic	Emphasized Text
5	<u>	Underlined text	Underlined Text
6	<s>	Strikethrough – no longer relevant	~~Strikethrough~~
7	<del>	Deleted text (editorial deletion)	Deleted Text

#	Tag	Description	Example / Output
8	<ins>	Inserted text (editorial insertion)	Inserted Text
9	<mark>	Highlighted / marked text	Highlighted Text
10	<small>	Smaller font-size text	Small Text
11	<sup>	Superscript text	E=mc2 (2 is superscript)
12	<sub>	Subscript text	H2O (2 is subscript)
13	<code>	Inline code snippet (monospace)	print("Hello")
14	<pre>	Preformatted text; preserves whitespace	Line 1 Line 2
15	<kbd>	Keyboard input by user	Ctrl + C
16	<samp>	Sample output from a program	Error: File not found
17	<var>	Variable in math or code	x + 5 = 10
18	<abbr>	Abbreviation with tooltip via title attr	HTML (full: HyperText Markup Language)
19	<cite>	Title of a creative work	The Great Gatsby
20	<dfn>	Defining instance of a term	Semantics: meaning in language
21	<q>	Short inline quotation (adds quotes)	"To be or not to be."
22	<blockquote>	Block-level quotation from external source	(indented block of quoted text)
23	<address>	Contact information for nearest section	contact@example.com
24	<time>	Machine-readable date/time value	January 1, 2025
25	<data>	Machine-readable data via value attribute	Product #998
26		Generic inline container for CSS/JS hooks	styled blue text
27	<bdi>	Bidirectional isolation for mixed text	Arabic text isolated
28	<bdo>	Bidirectional override (ltr/rtl)	dlroW olleH
29	<wbr>	Word break opportunity hint	longword / breakhere
30	<ruby>	Ruby annotation container (East Asian)	漢 with Kanji above
31	<rt>	Ruby annotation text (pronunciation)	にち above 日
32	<rp>	Ruby fallback parentheses	漢(kan)
33	<font>	⚠ Deprecated – use CSS instead	Deprecated in HTML5
34	<big>	⚠ Deprecated – use CSS font-size	Deprecated in HTML5
35		⚠ Deprecated – use CSS text-align	Deprecated in HTML5
36	<strike>	⚠ Deprecated – use <s> or <del>	Deprecated in HTML5

#	Tag	Description	Example / Output
37	<tt>	⚠ Deprecated – use <code>	Deprecated in HTML5

4. Conclusion / Analysis

This assignment provided a comprehensive understanding of all HTML text formatting tags from HTML 3.2 through HTML5. By creating a static web page with a structured table, it became clear how HTML tags serve two distinct purposes: semantic meaning and visual presentation.

A key takeaway is the distinction between semantic and physical tags. While <b> and both produce bold text visually, only conveys importance to browsers, screen readers, and search engines — making it the preferred choice for accessibility-compliant web development. Similarly, is preferred over <i> for stressed emphasis.

HTML5 deprecated several older tags such as <font>, <big>, , <strike>, and <tt>, replacing their functionality with CSS. This shift reflects the modern principle of separation of concerns — HTML for structure/semantics, CSS for presentation, and JavaScript for behaviour.

The assignment also highlighted the utility of specialised tags like <ruby> for East Asian typography, <bdi>/<bdo> for bidirectional text control in multilingual pages, and <time>/<data> for machine-readable metadata. These tags, though less commonly used, play a vital role in building accessible, internationally-compatible, and semantically rich web pages.

Overall, this exercise reinforced that a thoughtful use of HTML formatting tags leads to web pages that are not only visually effective but also accessible, search-engine friendly, and maintainable.